

- (e) Beskou 'n dun uniforme staaf met massa  $M$  en lengte  $L$ . Toon aan die traagheidsmoment om 'n as loodreg op die staaf en deur sy swaartepunt gegee word deur  $I = \frac{1}{12} ML^2$ .

*A thin uniform rod has a mass  $M$  and length  $L$ . Show that the moment of inertia about an axis which is perpendicular to the rod and which goes through its center of mass is given by  $I = \frac{1}{12} ML^2$ .*

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(5)

- (f) Wanneer die rotasie-as steeds loodreg op bogenoemde staaf is, maar deur die staaf se een endpoint gaan, bepaal die staaf se traagheidsmoment om hierdie as deur gebruik te maak van die teorie van parallelle asse.

*With the axis of rotation is still perpendicular to the rod mentioned above, but it goes through its one end, determine the moment of inertia about this axis by applying the parallel-axis theorem.*

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(3)

- (g) Elkeen van die drie lemme van 'n helikopter se rotor het 'n massa van 240 kg en is 5,2 m lank. Toon aan dat die traagheidsmoment van die rotor is  $I = 6,49 \times 10^3 \text{ kg m}^2$ .

*Each of the three helicopter rotor blades is 5,2 m long and has a mass of 240 kg. Show that the rotational inertia of the rotor is  $I = 6,49 \times 10^3 \text{ kg m}^2$ .*